PassFrame tuition

Percentages

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: Find 10% of 80.
Method & answer: 80 ÷ 10 = 8.

Standard: Find 30% of 150.
Method & answer: 10% of 150 = 15; 30% = 15 × 3 = 45.

Advanced: A jacket costs £60. It is reduced by 25% in a sale. What is the sale price?
Method & answer: 25% of £60 = £15. £60 − £15 = £45.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find 10% of 50.

2. What is 50% of 84? A) 21 B) 42 C) 48 D) 84

3. Find 25% of 40.

4. Find 10% of 230. Select ALL of the following that equal this value: A) 23 B) 2.3 C) 230 ÷ 10 D) 10% of 230 E) 32

5. Find 20% of 60.

6. Find 75% of 20.

7. What is 30% of 90? A) 18 B) 24 C) 27 D) 30

8. Find 100% of 45.

9. Find 5% of 200.

10. Find 40% of 25.

11. Find 10% of 96.

12. What is 50% of 37? A) 17 B) 18.5 C) 18 D) 37

13. Find 25% of 200. Select ALL of the following that equal this value: A) 50 B) 200 ÷ 4 C) A quarter of 200 D) 200 ÷ 2 E) 25

14. Find 60% of 50.

15. Find 90% of 10.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What is 15% of 80? A) 8 B) 10 C) 12 D) 15

2. Find 35% of 60.

3. A shirt costs £24. It is reduced by 25%. What is the new price?

4. A price of £50 increases by 20%. What is the new price?

6. What is 12% of 150? A) 12 B) 15 C) 18 D) 20

7. A TV costs £320 and is reduced by 15%. What is the sale price?

8. Out of 40 pupils, 6 are left-handed. Select ALL of the following that equal this percentage: A) 15% B) 6/40 simplified to 3/20 C) 0.15 D) 20% E) 3/20

9. Find 45% of 200.

11. Out of 80 sweets, 20 are red. What percentage is red?

12. What is 65% of 40? A) 20 B) 24 C) 26 D) 28

13. A book costing £15 is reduced by 20%. What is the new price?

14. Find 8% of 250.

5. In a class of 25, 40% are boys. How many boys are there?

10. A salary of £1,200 increases by 5%. What is the new salary?

15. Out of 60 pupils, 18 walk to school. Select ALL of the following that equal this percentage: A) 30% B) 18/60 simplified to 3/10 C) 0.3 D) 18% E) 3/10

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A laptop costs £400. It is first reduced by 10%, and then a further 10% off the new price. What is the final price? A) £320 B) £324 C) £360 D) £400

6. A car's value of £12,000 depreciates by 25% in the first year. What is its value after one year?

11. In a survey, 60% of 150 people liked football and the rest liked rugby. How many more people liked football than rugby?

2. A population of 2,000 grows by 15% in one year. What is the new population?

7. 45% of the 260 pupils in a school are girls. How many boys are there? A) 117 B) 130 C) 143 D) 234

12. A shop increases a £45 price by 10%, then reduces the new price by 10%. Select the correct statement. A) The final price is £45 (they cancel out) B) The final price is £44.55, less than £45 C) The final price is £49.50, more than £45 D) Cannot tell without a calculator

3. A shop's sales were £8,000 in January and rose to £9,200 in February. What was the percentage increase?

8. A price of £75 is reduced by 12%, then increased by 20% of the new price. What is the final price?

13. A charity's donations rose from £4,500 to £5,400. What was the percentage increase?

4. A jacket originally cost £80. In a sale it is reduced to £60. Select ALL statements that are TRUE: A) The discount is £20 B) The discount was 25% C) The discount was 20% D) The new price is 75% of the original E) The new price is 80% of the original

9. A test score improved from 40 out of 50 to 46 out of 50. Select ALL statements that are TRUE: A) The original score was 80% B) The new score is 92% C) The percentage increase in marks was 15% D) The percentage increase in marks was 6% E) The mark increased by 6 out of 50

14. Out of a school of 400 pupils, 35% travel by bus, 25% walk and the rest cycle. How many pupils cycle?

5. A number increases by 20% to become 96. What was the original number?

10. A number decreases by 30% to become 70. What was the original number?

15. A price is reduced by 40% to £54. What was the original price?